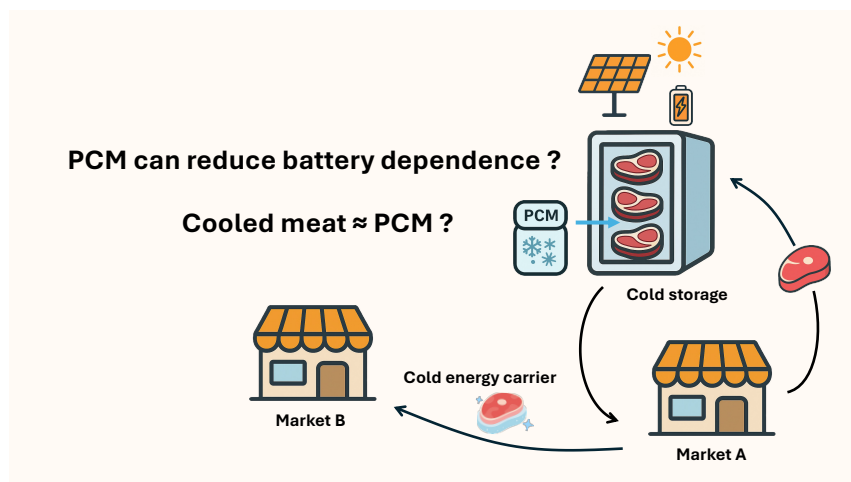


Graphical Abstract

Mobile Cold Energy Storage: Coupling Food Distribution and Energy Systems

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Highlights

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- A techno-economic framework for decentralized cold storage in informal markets
- Food products modeled as mobile carriers of cold energy in trade networks
- PCM thermal storage reduces system cost compared with battery storage
- Inter-market cold exchange lowers storage needs and system investment
- Coupling food logistics with energy systems improves cold-chain design

Mobile Cold Energy Storage: Coupling Food Distribution and Energy Systems

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ABSTRACT

Reliable cold storage remains a major constraint in informal food systems across sub-Saharan Africa. Perishable foods are mainly traded in open-air markets, where electricity access is often unstable or unavailable. Conventional cold-chain solutions that rely on continuous grid electricity or large battery systems are expensive and difficult to scale in these settings. This study develops a techno-economic framework to examine how cold energy can be generated, stored, and shared within informal food systems. We use open-air meat markets in Nigeria as a case study. The model integrates solar-powered refrigeration, phase change material (PCM) thermal storage, and inter-market food transport. It identifies least-cost combinations of energy supply, storage capacity and cold distribution strategies. A key feature of the model is that pre-chilled meat is treated as a mobile carrier of thermal energy, while PCM storage remains a stationary storage component within each market. This design links decentralised cold storage with existing trade routes and allows cold energy to move between markets through normal food transport. The results show that partial replacement of battery by PCM-based thermal storage can reduce total system cost by up to 15% compared with battery-only systems, primarily by substituting a significant share of electrical storage with stored cold energy and reducing battery capacity requirements. Sensitivity analysis shows that the PCM competitiveness depends on its capital cost, discharge efficiency, and operational flexibility. Allowing cold energy exchange between markets further reduces system investment (by up to 8%) by reallocating refrigeration and storage capacity across locations. Across all scenarios, the cost of cooling meat remains a small fraction of product value. This suggests that the proposed system can provide affordable cooling even under conservative assumptions. Beyond the Nigerian case, the modelling approach can be applied to other infrastructure-constrained food networks, where treating cold as an energy vector may enable more resilient and cost-effective cold-chain design.

1. Introduction

Nigeria, with over 200 million people, is the most populous country in Africa. Agriculture forms the backbone of its economy, accounting for over 24% of total gross domestic product (GDP) in 2020 and employing about two-thirds of the population [1]. Within this sector, livestock production plays a crucial role in food security and nutrition, with meat providing a significant share of dietary protein in Sub-Saharan Africa (SSA) [2, 3].

Across Nigeria and much of SSA, perishable foods are predominantly traded through informal open-air markets. These systems account for more than 80% of food sales and remain central to rural–urban food distribution [4, 5]. However, they operate with limited infrastructure, weak cold-chain capacity, and minimal electricity access [3, 6]. As a result, post-harvest losses are substantial. Nigeria loses approximately 40% of agricultural output annually, with associated income losses and greenhouse gas emissions [1, 7]. Informal meat markets are particularly exposed due to inadequate refrigeration, sanitation, and storage infrastructure [8].

Cold-chain systems based on vapor-compression refrigeration are effective in reducing spoilage [9, 10], but their deployment in informal markets is constrained by high capital costs and unreliable electricity supply [9, 11]. Solar-powered refrigeration offers an alternative in regions with high solar potential, including Nigeria [12, 13]. However, solar intermittency requires storage solutions to ensure continuous cooling. Battery-based storage is commonly used but significantly increases system cost and maintenance requirements [14]. Although recent studies demonstrate the

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technical feasibility of off-grid and hybrid solar cold storage systems [15, 16, 17], cost-effective and scalable storage strategies for informal markets remain unresolved.

Thermal energy storage using phase change materials (PCM) has been proposed as an alternative approach for stabilising refrigeration systems powered by intermittent renewable energy [18, 19]. By enabling cooling to be generated and stored locally without continuous electricity supply, PCM-based systems create distributed points of cold generation. However, existing approaches typically treat cooling infrastructure as a stationary energy system, independent of food trade dynamics [20]. In informal market networks, food moves continuously between spatially distributed markets. When food is cooled in markets equipped with PCM-based storage, the retained cooling travels with the product through existing trade routes. In this configuration, food acts as a mobile carrier of cold energy, while PCM remains stationary at the point of generation.

Despite growing interest in decentralised refrigeration, current studies do not explicitly model cold energy exchange within informal trade networks or quantify its system-level cost implications. To address this gap, this study develops a techno-economic optimisation framework to analyse how cold energy can be generated, stored, and exchanged across informal food systems. Using open-air meat markets (OAMM) in Nigeria as a case study, the framework integrates solar-powered refrigeration, PCM-based storage, and inter-market food transport within a unified optimisation model. By representing cooled food as a mobile carrier of thermal energy, the model captures interactions between decentralised energy infrastructure and trade flows, and determines least-cost configurations of generation, storage, and cold distribution strategies.

2. Methodology

This section describes the methodological framework used to evaluate decentralised cooling across OAMM in Nigeria. The approach combines field investigation with techno-economic optimisation modelling.

2.1. Field investigation and problem framing

Field investigations were conducted in August 2023 across five major open-air meat markets (OAMMs) in Abuja (Kubwa, Dei Dei, Orange, Kabusa and Lugbe). The selected sites represent variation in market sizes and spatial context. The objective was to characterise market structure, meat distribution patterns, refrigeration practices and electricity access conditions. Data collection included observations and informal interviews with meat vendors, which provided qualitative insights into operational routines, refrigeration methods, and cooperative behaviors among market participants. Particular attention was given to the multifunctional nature of OAMMs. An additional visit to a large open-air fish market (Kado) provided a reference case of shared cold storage powered by collectively operated diesel generators. These observations were used to define system boundaries and inform key modelling assumptions, including representative load profiles, daily operating schedules, cooling reliability requirements, and the feasibility of shared infrastructure at the market scale. Quantitative parameters (e.g., cooling demand, technology performance, and cost inputs) were derived from load simulations and published literature.

2.2. Method overview and scenario design

Based on the system characterisation, an hourly techno-economic optimisation model (Figure 1) was developed. The model determines least-cost configurations of solar photovoltaic (PV) generation, refrigeration capacity, and energy storage for decentralised cold-chain systems. Two storage technologies are considered: battery storage and phase change material (PCM)-based thermal storage integrated with refrigeration units. Three scenarios are evaluated:

- S1 (Battery-based decentralised systems): Each market operates an independent solar PV system with battery storage to ensure continuous cooling.
- S2 (PCM-based decentralised systems): PCM-based thermal storage is introduced at each market to substitute a large share of battery storage. PCM provides thermal buffering and reduces the system's reliance on electrical storage, while batteries are retained for short-term balancing.
- S3 (PCM with inter-market cold exchange): Building on S2, pre-chilled meat can be transported between markets. In this configuration, retained cooling embodied in the meat mass is modelled as a mobile thermal energy carrier. Cold-energy transfers between markets are represented as endogenous decision variables and optimised jointly with generation, storage, and refrigeration capacities.

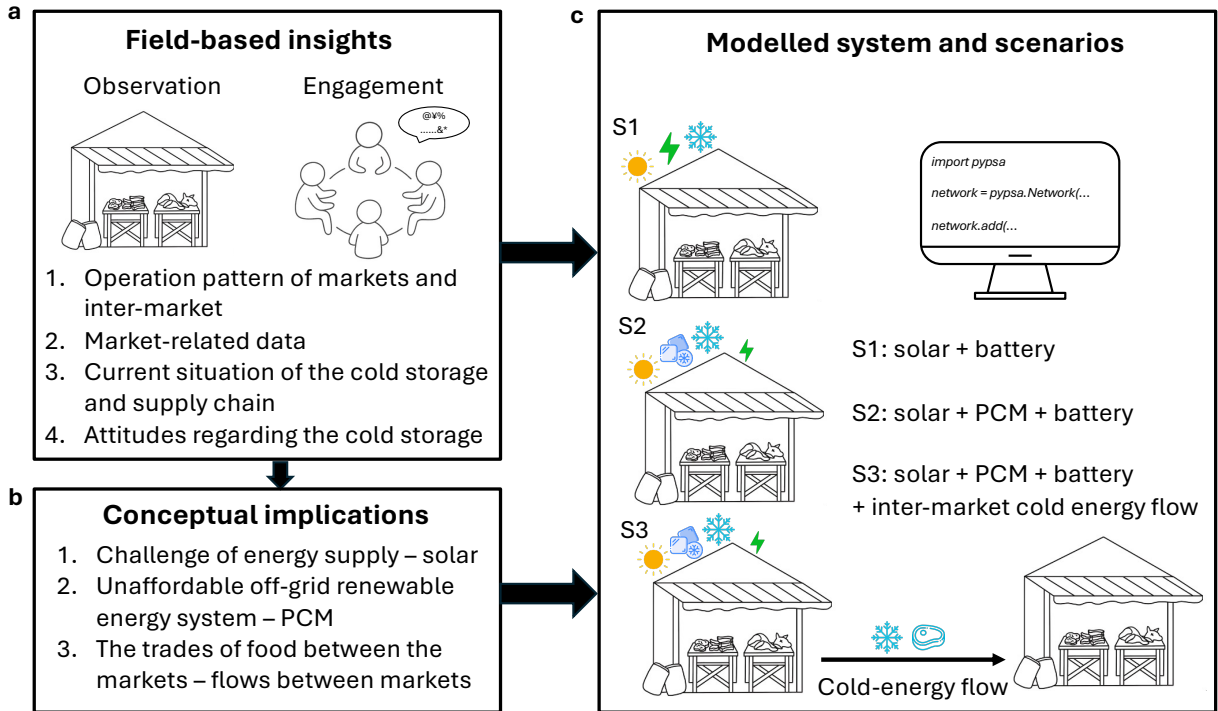


Figure 1: Overview of the modelling approach. (a) Field observations and stakeholder engagement used to define system boundaries and modelling assumptions; (b) key conceptual implications derived from field insights that motivate the representation of storage technologies and inter-market interactions; (c) the energy system model and scenario design, in which three configurations (S1–S3) are optimised using PyPSA.

Across all scenarios, system performance is evaluated using annualised total system cost as the primary metric. Cost components associated with generation, storage, and refrigeration are analysed separately. Sensitivity analysis is conducted on key cost and demand parameters to assess result robustness.

2.3. Model formulation and implementation in PyPSA

An hourly linear techno-economic optimisation model was formulated and implemented using the open-source Python for Power System Analysis (PyPSA) framework [21]. The model represents decentralised cold storage systems at the market scale, where each OAMM is modelled as a node comprising two energy carriers: electricity and cold (Figure 2). The electricity bus includes solar PV generation and, where applicable, battery storage. Electricity is converted into cooling via a refrigeration unit represented as a link with a fixed coefficient of performance (COP). Cooling demand is defined on the cold bus using simulated hourly load profiles described in the following section. PCM storage is connected to the cold bus and enables thermal energy to be stored and discharged independently of real-time electricity supply. Additional details of the optimisation formulation are provided in the SI Note 1.

The optimisation is performed at hourly resolution over the full year 2019. Solar resource data for Abuja (9.0643°N, 7.4893°E) are obtained from Renewables.ninja [22]. Given the close spatial proximity of the five markets, identical solar availability profiles are assumed. PV generation is represented using normalised hourly availability factors implemented through the PyPSA parameter p_{max_pu} and each market operates an independent PV installation. The model is formulated as a linear programming problem that minimises annualised total system cost, including investment in PV capacity, refrigeration units, battery storage (in S1), and PCM storage (in S2–S3). Investment decisions and hourly operational variables are optimised simultaneously. The optimisation is subject to electricity and cold energy balance constraints, storage state-of-charge dynamics, installed capacity limits, and full satisfaction of hourly cooling demand. Capital expenditures are annualised using standard annuity formulations based on technology lifetimes and a

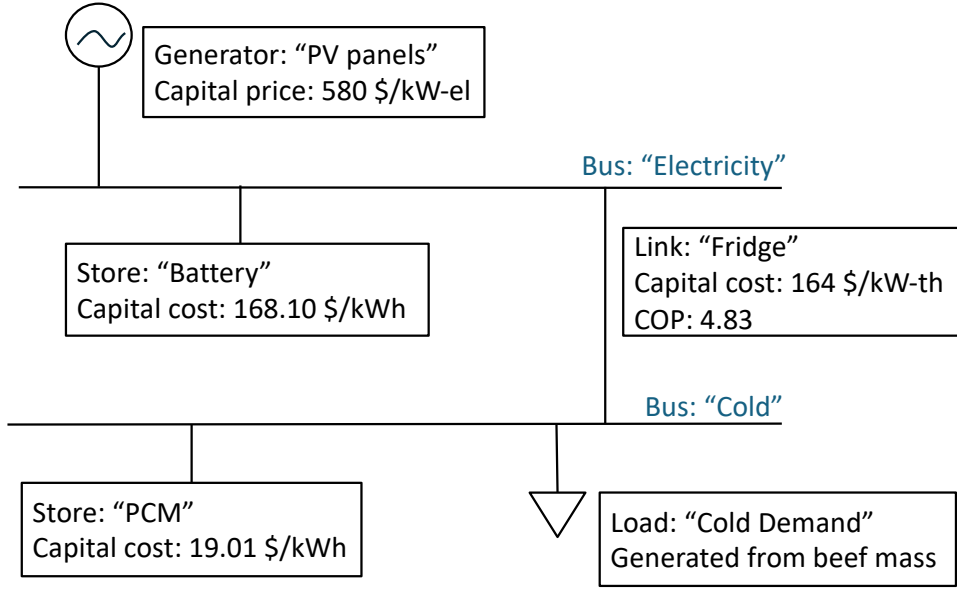


Figure 2: Outline of the cold storage energy system and the capital costs considered for each components (PV panels [23], battery storage [24], PCM storage [25] and refrigeration devices [26])

specified discount rate. Fixed and variable operation and maintenance costs are excluded due to their relatively small contribution to total cost in small-scale off-grid cold storage systems.

The three scenarios differ in storage configuration and network topology. In S1, each market operates a decentralised PV system with battery storage connected to the electricity bus. In S2, PCM-based thermal storage is connected to cold bus for partially substituting battery storage, reducing the need for electrical storage. In S3, inter-market links are introduced between cold buses to represent the transport of pre-chilled meat and the associated relocation of retained cooling. In this configuration, cooled meat is modelled as a mobile carrier of thermal energy, while PCM remains stationary at the point of cooling. Cold-energy transfers between markets are treated as endogenous decision variables and optimised jointly with generation, storage, and refrigeration capacities. Inter-market links are assumed to be lossless and unconstrained in capacity, representing an upper-bound estimate of the potential benefits of cold-energy exchange through existing trade networks. Transport costs are represented as marginal costs per unit of transferred cold energy. No additional electricity consumption for transport is modelled. Model outputs include least-cost system configurations, installed capacities of generation and storage technologies, hourly electricity and cold energy balances, and the spatial distribution of cold-energy flows in Scenario S3.

2.4. Refrigeration device demand modelling

Hourly refrigeration demand at each market was estimated using a first-principles thermal model based on sensible heat removal (Eq.1). The total daily cooling energy required to reduce meat temperature from an initial temperature ($T_{initial} = 40\text{ }^{\circ}\text{C}$) to the target storage temperature ($T_{target} = 4\text{ }^{\circ}\text{C}$)

$$E_{meat} = mc_p \Delta T \quad (1)$$

where $E_{meat}(t)$ is the total daily sensible cooling energy associated with the meat throughput, m the daily meat mass, c_p the specific heat capacity of beef (3.14 kJ/kgK is used). The initial temperature of 40 °C represents an upper-bound ambient condition for freshly handled meat in open-air markets.

Daily meat quantities were derived from vendor interviews across five OAMMs in Abuja. Animal counts were converted to meat mass using an average carcass yield of 275 kg per cattle [27]. The resulting daily cooling energy was distributed across 24 hours to generate time-resolved refrigeration demand profiles. Two operating regimes were defined: a daytime regime (04:00–19:00) representing slaughtering and active trading hours with elevated cooling

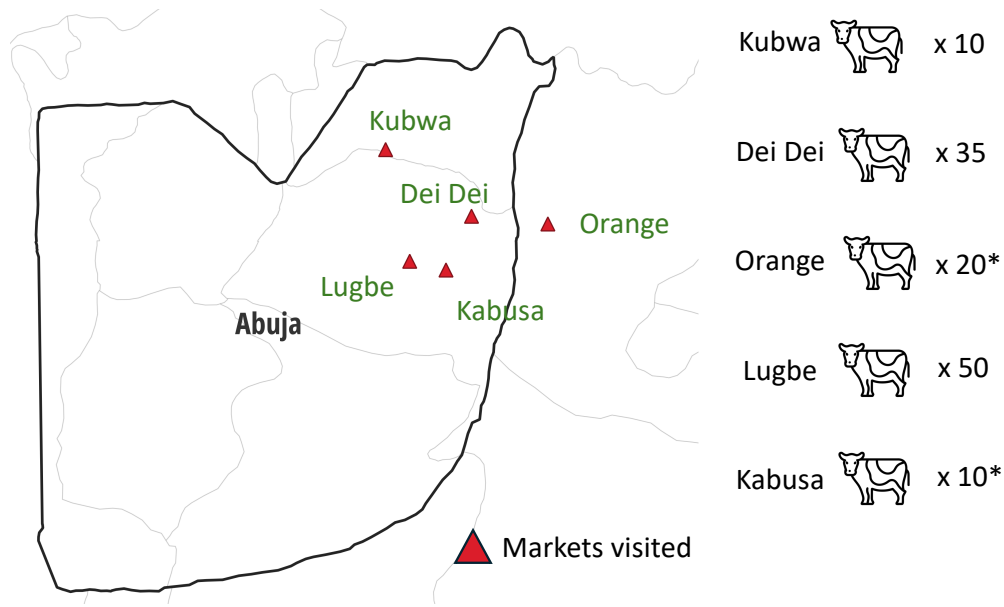


Figure 3: Five OAMMs visited in Abuja and the respective scale (* this value is estimated by the authors based on interviews with meat vendors)

demand, and a night-time regime (19:00–04:00) representing residual refrigeration of unsold meat. Within each regime, hourly load fractions were generated as non-negative and normalised values that preserve total daily cooling energy while introducing intra-day variability. The generated cooling demand profile is provided in SI Note 2.

The modelling approach assumes uniform product temperature during cooling, constant specific heat capacity, negligible microbial heat generation, and no additional external heat gains during the initial cooling process. These assumptions isolate the dominant first-order thermal process governing refrigeration demand in open-air market conditions.

2.5. Sensitivity analysis of thermal storage parameters

A sensitivity analysis was conducted to evaluate the influence of PCM storage parameters on system cost and technology allocation. Two parameters were varied: PCM capital cost and discharge efficiency. PCM capital cost was varied from \$0.75 to \$168.10 per kWh in \$1 increments, covering reported cost ranges and extending to the cost level of battery storage. Discharge efficiency, defined as the fraction of stored cold energy recoverable to meet demand, was varied from 0.1 to 1.0 in increments of 0.05. For each parameter combination, the optimisation model was solved for all five markets while holding all other inputs constant. No structural changes were introduced relative to the baseline formulation. The optimisation endogenously determines the least-cost allocation among PCM storage, battery storage, and refrigeration capacity. Outputs recorded for each run include total annual system cost and installed capacities of PCM and battery storage. Refrigeration capacity and runtime are additionally extracted to assess operational impacts associated with PCM charging requirements and changes in storage technology mix.

3. Results

3.1. Empirical characterisation of market-scale cooling constraints

Field investigations conducted across five open-air meat markets (OAMMs) in Abuja provide market-level operational data that are not captured in national statistics or publicly available datasets (the observation summary and photos are in SI Note 3 and Figure S2-6). In particular, quantitative information on daily meat throughput, operating hours, refrigeration practices, electricity access conditions, and inter-market redistribution patterns is largely absent from formal records. These empirical findings define key boundary conditions for decentralised cold-chain modelling. The five markets differ in daily cattle throughput and spatial location within the city (Figure 3), but exhibit consistent structural characteristics. All sites function as multifunctional nodes integrating livestock holding, slaughtering,



Figure 4: Multifunctional OAMM that includes keeping the animal, slaughtering, selling and redistribution

butchering, retail, and short-distance redistribution within a single physical space (Figure 4). Meat products are frequently transported between markets during trading hours, indicating that markets operate as interconnected nodes within a distributed trade network rather than as isolated retail endpoints.

Electricity access across all observed meat markets is intermittent and unreliable. Continuous mechanical refrigeration was not observed. Vendors primarily rely on ice blocks during daytime trading to slow spoilage under high ambient temperatures. In contrast, a large open-air fish market operated a shared cold room powered by a diesel generator, requiring collective fuel purchase and maintenance. Comparable fuel-based refrigeration systems were not present in meat markets, suggesting financial and operational barriers to continuous cooling technologies.

Cooperative practices related to energy use were observed at multiple sites. Traders routinely pool financial resources to purchase fuel or maintain shared equipment, demonstrating the existence of informal collective management mechanisms. Combined with frequent inter-market product redistribution, these findings indicate that cold-chain interventions must account for both decentralised operation at the market level and network-level interaction across markets.

Together, these empirical results establish three system-level constraints for cold-chain design in Nigeria's informal meat sector: (i) absence of continuous grid-based refrigeration, (ii) limited access to capital-intensive storage technologies, and (iii) existing spatial linkages between markets through routine product redistribution. These constraints motivate the decentralised and network-integrated modelling framework evaluated in the subsequent analysis.

3.2. Cost comparison between battery-based and PCM-based cold storage

Across the five markets, integrating PCM-based thermal storage (S2) reduces total annualised system cost up to 15%, with an average reduction of 11% relative to the battery-only configuration (S1). This cost reduction is primarily

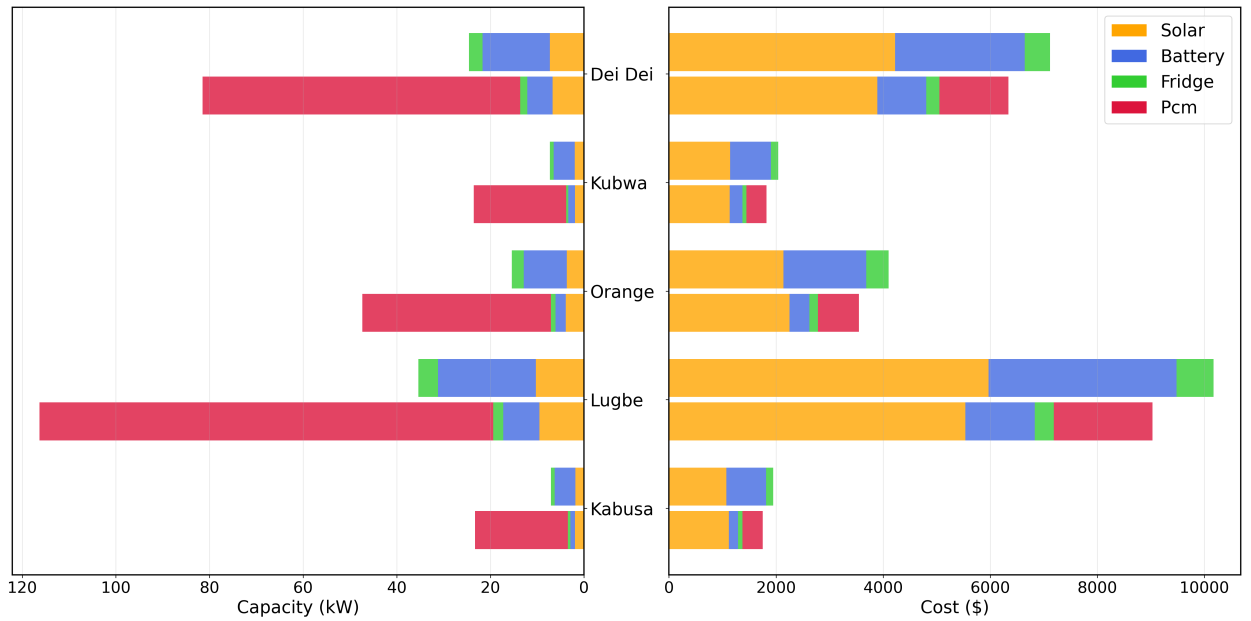


Figure 5: Capacity and cost comparison between S1 and S2

driven by substantial decreases in required battery capacity. Figure 5 shows the optimised capacities under both scenarios. When PCM storage is introduced, installed battery capacity decreases by approximately 67% on average across markets. PV capacity increases modestly (around 4%) to accommodate daytime charging of thermal storage. Installed refrigeration capacity is reduced by roughly 50%, reflecting a shift from continuous high-power cooling toward load shifting through stored cold energy. PCM substitutes for a substantial share of electrical storage while batteries remain necessary for balancing. Although S2 requires additional investment in thermal storage and slightly larger PV arrays, these costs are more than offset by the reduction in battery capacity, resulting in lower total system cost. Hourly operation patterns illustrate the underlying mechanism (Figure 6). In S1, early morning and evening cooling demand is supplied primarily through battery discharge due to the temporal mismatch between solar generation and refrigeration demand. This results in deep daily battery cycling. In S2, part of the daytime solar output is diverted to charge PCM storage, which is later discharged to meet evening cooling demand. Battery operation is consequently reduced and limited mainly to short-term balancing. The system thus relies more on stored thermal energy and less on electrical storage to manage solar intermittency.

3.3. Inter-market cold energy flows and system cost outcomes

Allowing inter-market cold energy exchange (S3) reduces total system cost by 8% relative to the isolated PCM-based configuration (S2) (Table 1). The cost reduction is primarily driven by a 35% decrease in required PCM storage capacity at the system level, while total PV capacity remains nearly unchanged. The introduction of cold-energy transfer enables spatial reallocation of cooling resources across markets. Rather than sizing PCM storage independently at each site to meet local peak demand, the optimisation exploits cross-market exchange to balance temporal mismatches collectively. As a result, aggregate storage requirements decline without requiring additional generation capacity.

At the individual market level, cost impacts are heterogeneous. Kabusa and Orange experience the largest cost reductions (4% each), accompanied by decreases in both PV and PCM capacities. Lugbe and Dei Dei show modest cost reductions with relatively small capacity adjustments. In contrast, Kubwa exhibits a slight cost increase (+3%), associated with higher PV and refrigeration capacities despite reduced PCM storage. This indicates that Kubwa functions primarily as a net exporter of cold energy within the optimised network, investing in additional generation and refrigeration capacity to supply other markets.

Export and import shares increase across all markets once transfer is enabled, confirming active utilisation of inter-market cold flows. These results demonstrate that cold exchange allows decentralised systems to operate as a coordinated network, reducing aggregate storage requirements through spatial diversification of cooling demand.

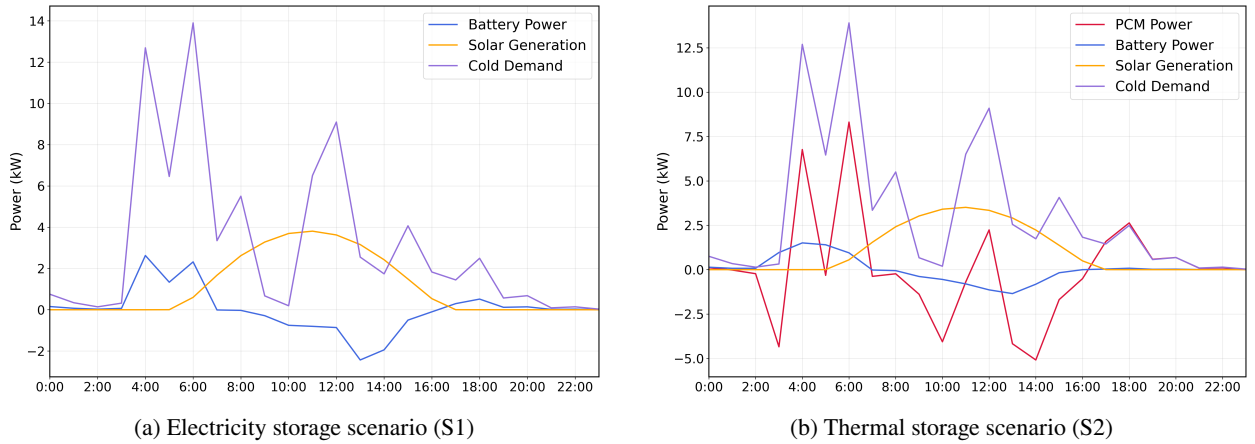


Figure 6: Cold demand and supply alignment comparison

Table 1

Relative changes (%) in system cost and capacity decisions when enabling inter-market cold energy transfer via chilled meat, compared with S2 (negative values indicate reductions)

Market	Total Cost	PV Capacity	Battery Cost	PCM Capacity	Fridge Capacity	Export Percentage	Import Percentage
Kabusa	-4	-3	15	-16	-9	23	39
Lugbe	-1	1	-12	-2	6	10	7
Orange	-4	-3	6	-12	-6	25	39
Kubwa	3	6	3	-11	11	34	12
Dei Dei	-2	-2	-18	6	8	16	13
Total	-8	0	-6	-35	11	107	110

3.4. Sensitivity of system cost and storage allocation to PCM parameters

Sensitivity analysis is performed on the decentralised PCM-based configuration (S2) to isolate storage substitution dynamics without confounding effects from inter-market cold exchange. By holding network topology constant, the analysis characterises how PCM capital cost and discharge efficiency jointly determine storage allocation under decentralised operation. Across all five markets, PCM competitiveness depends on both its relative capital cost and discharge efficiency. A clear transition frontier separates PCM-dominated solutions from battery-dominated solutions in the cost–efficiency parameter space (Figure 7). PCM is selected when it is sufficiently inexpensive and exhibits moderate-to-high discharge efficiency, whereas battery storage dominates as the PCM-to-battery cost ratio increases or discharge efficiency declines. The position of this frontier varies across markets. Lugbe, the largest market, exhibits the broadest PCM-preferred region, indicating that higher cooling demand improves the economic viability of thermal storage. Orange and Dei Dei display intermediate behaviour, while Kabusa and Kubwa show narrower PCM-preferred regions, with PCM selected only under favourable cost and efficiency conditions. These results suggest that market scale influences storage substitution patterns, with larger markets better able to exploit thermal storage economies.

To contextualise the model-derived frontier, reported properties of two representative PCM materials commonly used in refrigeration applications are summarised in Table 2. The selected materials, n-tetradecane (organic PCM) and sodium sulfate decahydrate (inorganic PCM), are used as illustrative benchmarks spanning typical ranges of latent heat, density, discharge efficiency, and capital cost reported in the literature. Figure 8 positions these materials within the cost–efficiency parameter space. Both materials occupy a discharge efficiency range of 0.7–0.9 and correspond to PCM-to-battery cost ratios of approximately 0.11–0.23. Within this range, system cost increases markedly as the cost ratio rises, indicating that relative capital cost exerts a stronger influence on competitiveness than variations within the considered efficiency band. Sodium sulfate decahydrate, positioned at lower cost ratios (0.11–0.13), lies closer to the

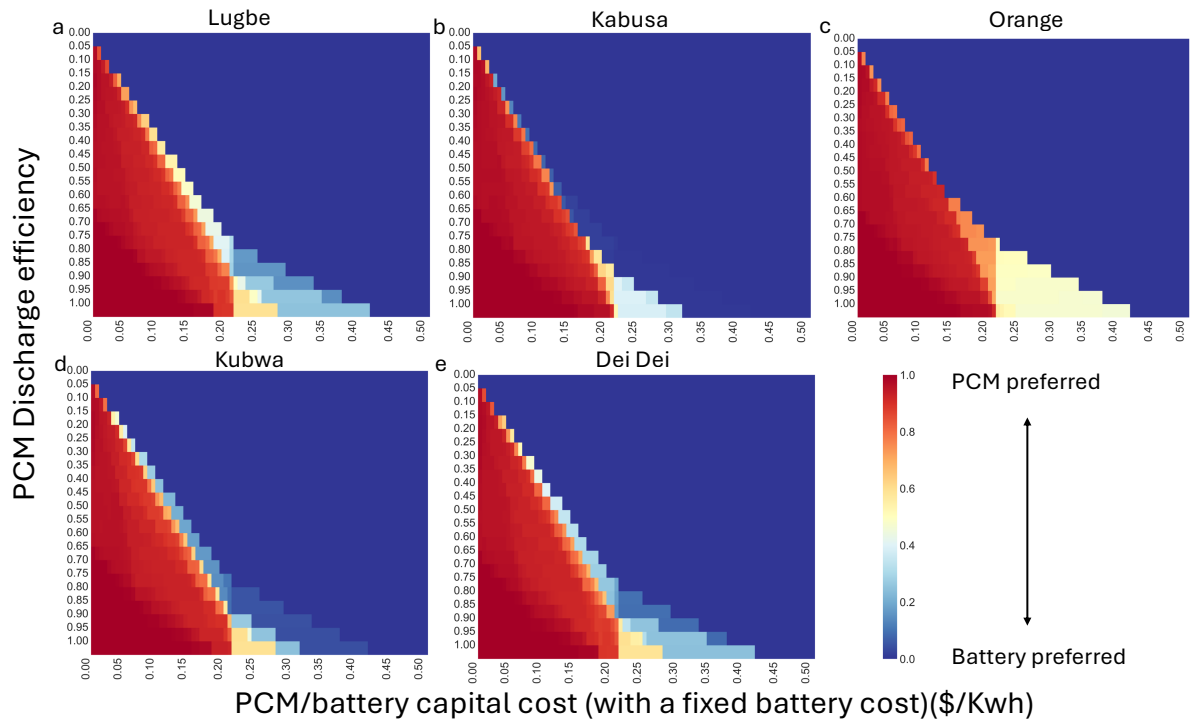


Figure 7: Effect of PCM efficiency and price on the choice between PCM and battery storage.

Table 2

Properties and cost parameters of selected PCMs [25]

Type	Phase Change Temperature (°C)	Density (kg/m ³)	Latent Heat (kJ/kg)	Discharge Efficiency	Unit Price (\$/ton)	Capital Cost (\$/kWh)
n-tetradecane (C14)	5	767	250	0.7–0.9	1980–2640	28.51–38.02
Sodium sulfate decahydrate	7.5	1350	150	0.7–0.9	792–924	19.01–22.18

minimum-cost envelope than n-tetradecane (0.17–0.23), suggesting greater economic viability under the assumed cost parameters.

Refrigeration charging time further shapes storage allocation (Figure 9). Total system cost decreases as the available charging window increases. Short charging durations (below approximately 3–4 hours) correspond to battery-dominated solutions, as meeting daily cooling demand within a narrow time window requires high instantaneous charging power. In these cases, batteries provide short-timescale flexibility without necessitating oversized refrigeration capacity. As charging time increases, peak power requirements decline, enabling a larger share of storage to be provided by PCM, reflected in rising PCM-to-battery capacity ratios and lower overall system cost.

4. Discussion

This study identifies two linked substitution mechanisms in decentralised cold chains. First, PCM substitutes for part of the battery function by shifting cooling across time. Second, once chilled food moves between markets, part of the storage function itself becomes mobile, allowing cold generated in one market to serve demand in another. In the Abuja case, the PCM-based configuration reduces annualised system cost by up to 15%, with an average reduction of 11%, mainly through a roughly 67% reduction in battery capacity. Allowing inter-market cold exchange reduces

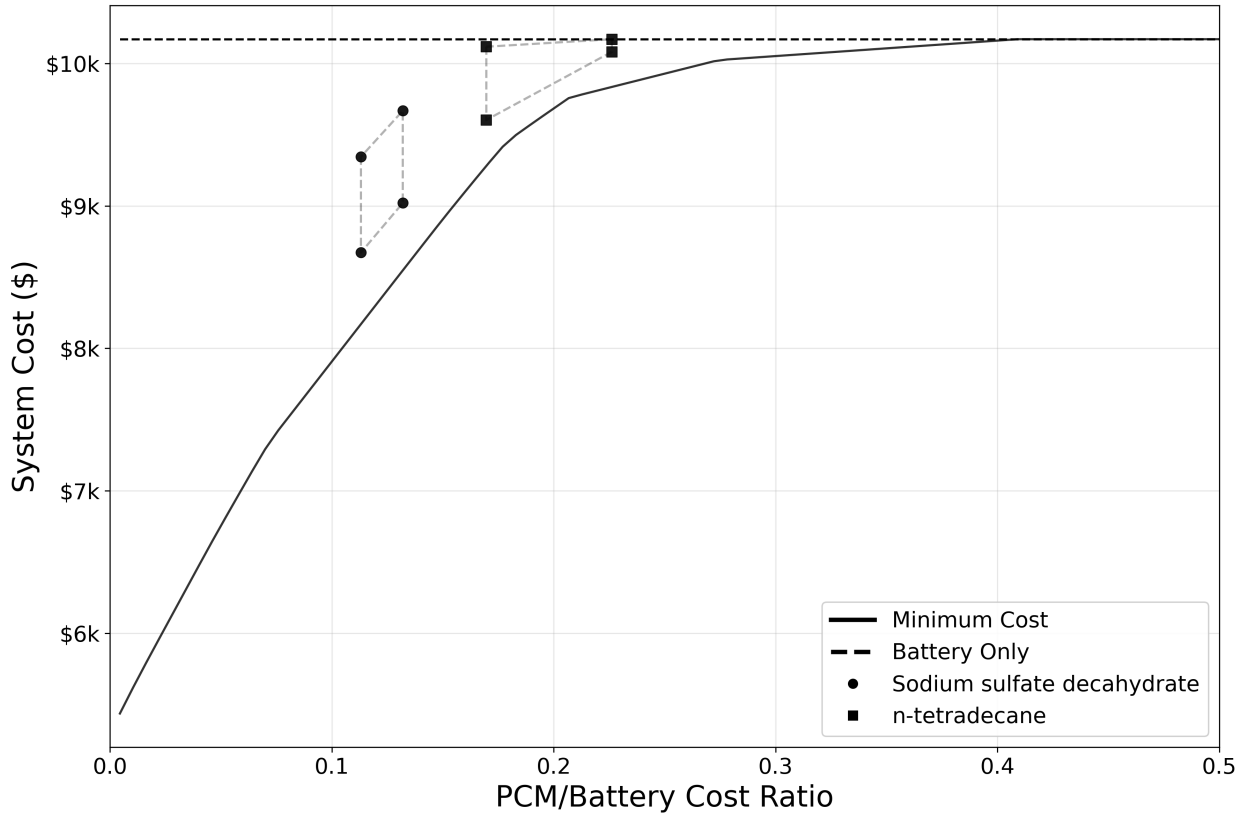


Figure 8: Benchmarking real PCM candidates against the model-derived minimum-cost envelope in Lugbe. The solid curve shows the minimum system cost across the PCM cost–efficiency parameter sweep, while the dashed line indicates the battery-only benchmark. Markers represent reported cost and discharge efficiency ranges for sodium sulfate decahydrate and n-tetradecane (C14).

total system cost by a further 8% and aggregate PCM capacity by 35% relative to the isolated PCM case. The main contribution is therefore not only a comparison of storage technologies, but a demonstration that cold-chain design in informal markets depends on how storage is allocated across both time and space.

4.1. Substitution between batteries and PCM in markets

The model shows that PCM becomes favourable under three conditions: low relative capital cost, moderate to high discharge efficiency, and sufficient charging time. In the sensitivity analysis, a clear frontier separates PCM-dominated from battery-dominated solutions. PCM is selected when it is inexpensive enough relative to batteries and can recover a large enough fraction of stored cold; battery storage becomes dominant as PCM cost rises or discharge efficiency falls. The results also show that market scale matters. Lugbe, the largest market, has the broadest PCM-preferred region, while smaller markets such as Kabusa and Kubwa only favour PCM under more restricted cost and efficiency conditions. Charging duration is equally important. When refrigeration must charge within a short window, around 3 to 4 hours or less, the model shifts toward battery-dominated solutions because batteries provide the high instantaneous flexibility needed to meet demand without oversizing refrigeration. As the charging window lengthens, PCM takes a larger role and total system cost falls. The mechanism is therefore specific: PCM is most valuable when daily cooling demand is predictable and can be pre-produced over a long enough daytime window, while batteries remain valuable when flexibility must be delivered quickly and under tighter operating schedules.

4.2. Substitution through mobile cold storage

The inter-market mechanism only works when three conditions are present: routine product redistribution between nearby markets, unequal local cooling needs across the network, and at least some markets that can economically

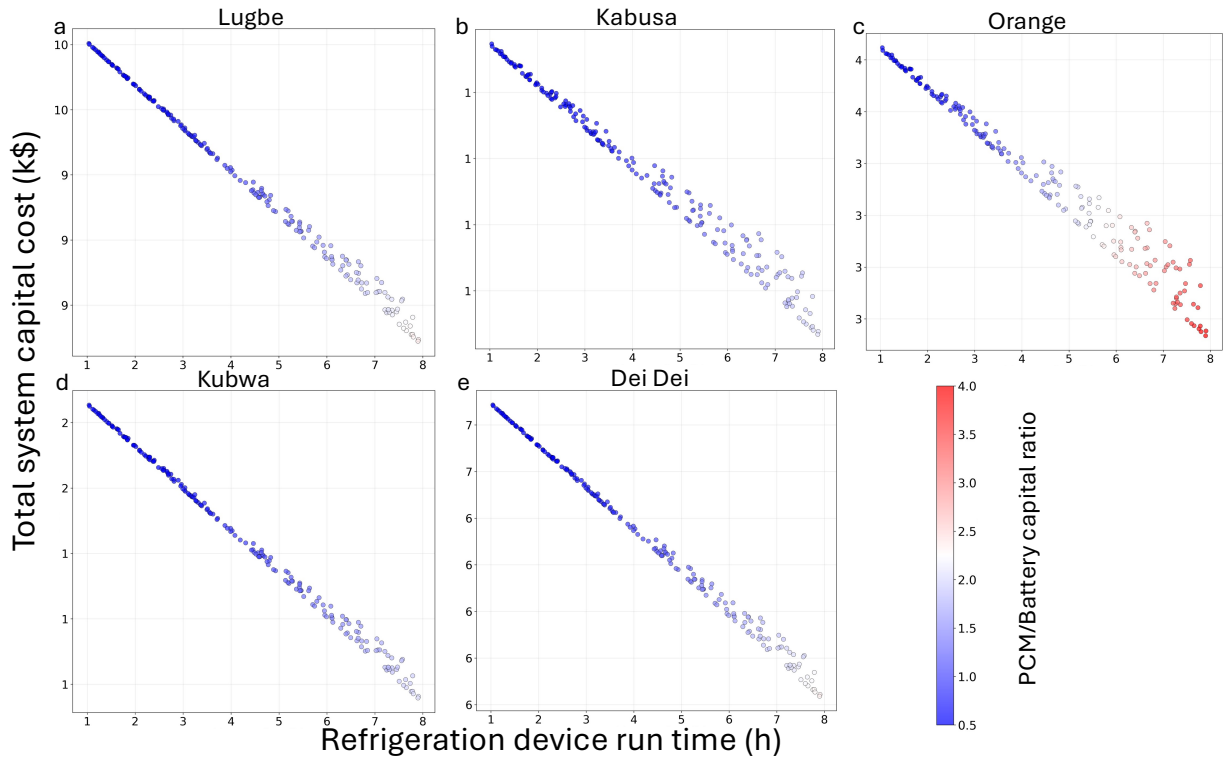


Figure 9: Effect of refrigeration-device runtime and PCM capacity on total system cost.

export retained cold. Under these conditions, markets no longer need to size PCM independently against local peaks. Instead, part of the storage requirement is shifted into trade flows, which is why allowing cold exchange cuts aggregate PCM capacity by 35% while leaving total PV capacity almost unchanged. This shows that the main benefit of mobile cold exchange is not more generation, but better spatial allocation of storage and refrigeration resources. The benefits are also uneven. Kabusa and Orange see the largest cost reductions, around 4%, while Lugbe and Dei Dei gain only modestly. Kubwa, by contrast, experiences a slight cost increase of 3% because it takes on a net exporter role and invests in more PV and refrigeration capacity to support the network. This suggests that mobile cold storage is most valuable in clustered local market systems with existing short-distance redistribution and asymmetric market roles, especially where some nodes can act as cooling hubs rather than every market being built as a fully independent system.

4.3. Implications for cold-chain design and policy

These results support a shift from isolated-market planning to cluster-based cold-chain design. The model does not suggest removing batteries altogether. Rather, it indicates a more differentiated design logic: PCM can provide the main low-cost storage function where cooling demand is regular and charging windows are long, while batteries remain useful for short-timescale balancing and constrained operating schedules. For policy, this means off-grid cold storage programmes should not default to battery-led designs when the service being provided is predictable cooling rather than general electricity flexibility. The results also suggest that investment should target groups of connected markets rather than single sites in isolation. Because the gains in the inter-market case come mainly from reallocating storage rather than expanding PV, effective policy will depend as much on route structure, handling rules, and temperature governance as on the hardware itself. In the Nigerian and wider African context, where informal food systems already rely on short-distance redistribution and shared practices, these networks should be treated not only as infrastructure-poor systems but also as coordination structures that can support more distributed forms of cold-chain development.

4.4. Effects on vendors, market dynamics, and stakeholders

The two mechanisms identified here imply different changes for vendors and market actors. The temporal substitution mechanism mainly reduces dependence on battery investment, which lowers the capital burden of decentralised cooling and may make shared cold storage more financially accessible to trader groups or market associations. The spatial substitution mechanism has a more uneven effect. Because some markets can become net exporters of cold while others mainly import it, the benefits of the network are not distributed equally. In this study, Kubwa takes on extra investment to support exchange, while markets such as Kabusa and Orange gain larger cost reductions. In practice, this could reshape bargaining relationships between hub markets, smaller dependent markets, transporters, and storage operators. Vendors may benefit from lower spoilage risk, more flexible selling times, and better inventory control, but they may also become more dependent on whoever controls shared cooling assets or access to exported cold. Since field observations already show both inter-market redistribution and collective resource-sharing practices, the institutional question is not whether coordination is possible, but how costs, access, and decision-making are organised. Once cold storage becomes partly mobile, the system is no longer only a technical infrastructure problem; it also becomes a question of governance, pricing, and trust across the market network.

4.5. Limitations and future work

Several limits should qualify the current findings. The inter-market case is best understood as an upper-bound estimate because transfers are modelled in simplified form, without explicit transport losses, route constraints, or vehicle energy use. Cooling demand is also simplified around sensible heat removal for a single product type and does not explicitly capture detailed quality decay, microbial processes, or broader food heterogeneity. In addition, the empirical base is limited to a small Abuja market network, and the model does not endogenise financing, ownership, maintenance capacity, or stakeholder behaviour. These limits do not remove the core insight, but they do mean that the quantitative savings should be interpreted as structural potential rather than deployment-ready outcomes.

Future work should therefore extend the framework in four directions. First, it should include multiple food types as distinct thermal vectors, since meat, fish, dairy, and produce differ in heat capacity, spoilage dynamics, and transport value. Second, it should test the concept at larger regional scales with longer routes, more diverse climates, and more complex market hierarchies. Third, it should integrate explicit transport losses, travel times, and time-temperature quality dynamics. Fourth, it should examine ownership models, pricing rules, and cooperative governance through field-based pilots. More broadly, the concept developed here opens a wider research agenda on material-embedded energy storage in infrastructure-constrained systems, where useful energy may travel not only through wires and fuels, but also through goods moving within economic networks.

5. Conclusion

This study develops a techno-economic optimisation framework to evaluate decentralised solar-powered cold storage systems in Nigeria's informal meat markets. By integrating field observations, refrigeration demand modelling, and energy system optimisation, the framework analyses how solar generation, storage technologies, and food logistics jointly determine least-cost cold-chain configurations.

The results reveal three key findings. First, thermal storage using PCM can substitute for a significant share of electrical storage by shifting cooling provision across time, reducing battery capacity requirements and lowering system cost under favourable cost conditions. Second, operational flexibility—represented by allowable refrigeration charging duration—strongly influences storage composition, with longer charging windows favouring PCM-based storage and shorter windows favouring battery-based solutions. Third, enabling inter-market cold exchange introduces spatial coordination into decentralised cold systems. Embodied cold energy transported through food distribution networks reduces aggregate PCM capacity requirements and lowers total system cost, demonstrating that mobile cold flows can function as distributed thermal storage.

More broadly, the findings highlight how decentralised food distribution networks can be integrated with renewable energy systems through thermal storage and embodied energy flows. By representing chilled products as mobile carriers of cold energy, the framework extends conventional energy system modelling beyond infrastructure-bound energy carriers and provides a basis for analysing coordinated cold-chain design in infrastructure-constrained food systems.

CRedit authorship contribution statement

Hange Lao: Conceptualization, Methodology, Data curation, Software, Visualization, Writing - Original draft preparation, Writing - review&editing. **Binjian Nie:** Data curation, Writing - review. **Wei He:** Conceptualization, Methodology, Supervision, Writing - review&editing.

References

- [1] D. Onwude, T. Motmans, K. Shoji, R. Evangelista, J. Gajardo, D. Odion, N. Ikegwuonu, O. Adekanmbi, S. Hourri, T. Defraeye, Bottlenecks in nigeria's fresh food supply chain: What is the way forward?, *Trends in Food Science & Technology* 137 (2023) 55–62.
- [2] N. W. Smith, A. J. Fletcher, J. P. Hill, W. C. McNabb, Modeling the contribution of meat to global nutrient availability, *Frontiers in Nutrition* 9 (2022) 766796.
- [3] K. Roesel, D. Grace, Food safety and informal markets: Animal products in sub-Saharan Africa, Routledge, 2014.
- [4] D. Grace, K. Makita, E. Kang'ethe, B. Bonfoh, K. Roesel, Taking food safety to informal markets, in: *Food Safety and Informal Markets*, Routledge, 2014, pp. 11–22.
- [5] C. Hannah, J. Davies, R. Green, A. Zimmer, P. Anderson, J. Battersby, K. Baylis, N. Joshi, T. P. Evans, Persistence of open-air markets in the food systems of africa's secondary cities, *Cities* 124 (2022) 103608.
- [6] R. O. Adesola, D. Hossain, O. A. Ogundijo, I. Idris, A. Hamzat, B. H. Gulumbe, A. A. Bakre, O. G. Banwo, D. E. Lucero-Priso III, Challenges, health risks and recommendations on meat handling practices in africa: A comprehensive review, *Environmental health insights* 18 (2024) 11786302241301991.
- [7] G. Seth, Nigeria: Food smart country diagnostic, <https://documents1.worldbank.org/curated/en/703791601302657183/pdf/Nigeria-Food-Smart-Country-Diagnostic.pdf> (2020).
- [8] H. K. Adesokan, O. C. Obimdike, V. O. Adetunji, Informal and formal meat marketing in ibadan, nigeria: public health implications from microbial assessment, *PAMJ-One Health* 5 (15) (2021).
- [9] W. Amjad, A. Munir, F. Akram, A. Parmar, M. Precoppe, F. Asghar, F. Mahmood, Decentralized solar-powered cooling systems for fresh fruit and vegetables to reduce post-harvest losses in developing regions: a review, *Clean Energy* 7 (3) (2023) 635–653.
- [10] F. Yamauchi, H. Takeshima, J. Estudillo, Y. Kijima, T. Sonobe, Solar power to transform food systems and rural livelihoods: evidence from a solar-powered cold storage intervention in nigeria, *Agricultural Development in Asia and Africa: Emerging-economy State and International Policy Studies*. Singapore: Springer (2023) 329–341.
- [11] M. Xue, J. Zhang, W. Tang, Optimal temperature control for quality of perishable foods, *ISA transactions* 53 (2) (2014) 542–546.
- [12] IEA, Africa energy outlook 2022, <https://www.iea.org/reports/africa-energy-outlook-2022> (2022).
- [13] M. Shaaban, J. Petinrin, Renewable energy potentials in nigeria: Meeting rural energy needs, *Renewable and sustainable energy reviews* 29 (2014) 72–84.
- [14] H. Amini Toosi, Life cycle cost optimization of battery energy storage systems for bipv-supported smart buildings: A techno-economic analysis, *Sustainability* 17 (13) (2025) 5820.
- [15] K. Mehta, B. Lwakatare, W. Zörner, M. Ehrenwirth, Mini-grid performance in sub-saharan africa: case studies from tsumkwe and gam, namibia, *Sustainable Energy Research* 12 (1) (2025) 30.
- [16] J. Meyer, S. von Solms, Design considerations for reducing battery storage in off-grid, stand-alone, photovoltaic-powered cold storage in rural applications, *Energies* 15 (9) (2022) 3468.
- [17] T. Ayodele, T. Mosethle, A. Yusuff, A. Ogunjuigbe, Off-grid hybrid renewable energy system with hydrogen storage for south african rural community health clinic, *International Journal of Hydrogen Energy* 46 (38) (2021) 19871–19885.
- [18] H. Selvnnes, Y. Allouche, R. I. Manescu, A. Hafner, Review on cold thermal energy storage applied to refrigeration systems using phase change materials, *Thermal Science and Engineering Progress* 22 (2021) 100807.
- [19] S. Bista, S. E. Hosseini, E. Owens, G. Phillips, Performance improvement and energy consumption reduction in refrigeration systems using phase change material (pcm), *Applied Thermal Engineering* 142 (2018) 723–735.
- [20] P. Iyer, D. Robb, Cold chain optimisation models: A systematic literature review, *Computers & Industrial Engineering* 204 (2025) 110972.
- [21] T. Brown, J. Hörsch, D. Schlachtberger, PyPSA: Python for Power System Analysis, *Journal of Open Research Software* 6 (4) (2018). arXiv:1707.09913, doi:10.5334/jors.188. URL <https://doi.org/10.5334/jors.188>
- [22] S. Pfenninger, I. Staffell, Long-term patterns of european pv output using 30 years of validated hourly reanalysis and satellite data, *Energy* 114 (2016) 1251–1265.
- [23] H. Takeshima, F. Yamauchi, H. O. Edeh, M. A. Hernandez, Solar-powered cold-storage and agrifood market modernization in nigeria, *Agricultural Economics* 54 (2) (2023) 234–255.
- [24] T. Brown, D. Schlachtberger, A. Kies, S. Schramm, M. Greiner, Synergies of sector coupling and transmission reinforcement in a cost-optimised, highly renewable european energy system, *Energy* 160 (2018) 720–739.
- [25] S. Tong, B. Nie, Z. Li, C. Li, B. Zou, L. Jiang, Y. Jin, Y. Ding, A phase change material (pcm) based passively cooled container for integrated road-rail cold chain transportation—an experimental study, *Applied Thermal Engineering* 195 (2021) 117204.
- [26] A. R. Razmi, A. Arabkoohsar, H. Nami, Thermoeconomic analysis and multi-objective optimization of a novel hybrid absorption/recompression refrigeration system, *Energy* 210 (2020) 118559.
- [27] Á. Vaskó, I. Vida, L. Vasa, F. Adzitey, Opportunities within the meat supply chain in africa—the case of beef production in northern ghana, *PloS one* 17 (1) (2022) e0260668.